

HDOC - Day 9

Menu-Driven C Programs

Question 1 - Swap Two Numbers

Input: two numbers a and b. **Output:** values before and after swapping. Option 1 uses a third variable; option 2 swaps without a third variable.

Algorithm

1. Start. 2. Read a and b. 3. Display the menu. 4. Read choice. 5. If choice 1, use temp=a, a=b, b=temp. 6. If choice 2, use a=a+b, b=a-b, a=a-b. 7. Display values before and after swapping. 8. Repeat until Exit. 9. Stop.

Pseudocode

```
READ a,b
REPEAT
  DISPLAY menu; READ choice
  CASE 1: temp=a; a=b; b=temp
  CASE 2: a=a+b; b=a-b; a=a-b
  CASE 3: exit
  OTHERWISE: print "Invalid choice"
UNTIL choice=3
```

Flowchart

START -> Input a,b -> Menu -> Read choice -> [1: third-variable swap / 2: arithmetic swap / 3: Exit / invalid] -> Display result -> Menu -> STOP.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	12,7; choice 1	After: 7,12	After: 7,12	Pass
2	25,40; choice 2	After: 40,25	After: 40,25	Pass
3	choice 9	Invalid choice	Invalid choice	Pass

C Program

```
#include <stdio.h>
int main() {
    int a,b,temp,choice;
    printf("Enter two numbers: "); scanf("%d%d",&a,&b);
    do {
        printf("\n1. Swap using third variable\n2. Swap without third variable\n3. Exit\nEnter choice: ");
        scanf("%d",&choice);
        switch(choice) {
            case 1: printf("Before swapping: a = %d, b = %d\n",a,b);
                    temp=a; a=b; b=temp;
                    printf("After swapping : a = %d, b = %d\n",a,b); break;
            case 2: printf("Before swapping: a = %d, b = %d\n",a,b);
                    a=a+b; b=a-b; a=a-b;
                    printf("After swapping : a = %d, b = %d\n",a,b); break;
            case 3: break;
            default: printf("Invalid choice.\n");
        }
    } while(choice!=3);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang swap.c -o swap
puppie@Anvikas-MacBook-Air Desktop % ./swap
Enter two numbers: 12 7
1. Swap using third variable
2. Swap without third variable
3. Exit
Enter choice: 1
Before swapping: a = 12, b = 7
After swapping : a = 7, b = 12
```

```
puppie@Anvikas-MacBook-Air Desktop % ./swap
Enter two numbers: 25 40
Enter choice: 2
Before swapping: a = 25, b = 40
After swapping : a = 40, b = 25
```

Result: The swapping program was compiled, executed and tested successfully.

Question 2 - Sum of Natural Numbers and Sum of Squares

Input: positive integer n. **Output:** sum of first n natural numbers or sum of their squares, according to the menu choice.

Algorithm

1. Start. 2. Read positive n. 3. Display menu. 4. For option 1, add i from 1 to n. 5. For option 2, add i^2 from 1 to n. 6. Display result. 7. Repeat until Exit. 8. Stop.

Pseudocode

```
READ n
REPEAT
  DISPLAY menu; READ choice
  IF choice=1: sum=0; FOR i=1 TO n: sum=sum+i
  IF choice=2: sum=0; FOR i=1 TO n: sum=sum+i*i
  DISPLAY result
UNTIL choice=3
```

Flowchart

START -> Input n -> Menu -> [1: add 1..n / 2: add squares / 3: Exit / invalid] -> Display result -> Menu -> STOP.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n=5; choice 1	15	15	Pass
2	n=5; choice 2	55	55	Pass
3	n=1; choice 2	1	1	Pass

C Program

```
#include <stdio.h>
int main() {
    int n,choice,i; long sum;
    do { printf("Enter n: "); scanf("%d",&n); } while(n<=0);
    do {
        printf("\n1. Sum of first n natural numbers\n2. Sum of squares of first n natural numbers\n3. Exit\nEnter choice: ");
        scanf("%d",&choice);
        switch(choice) {
            case 1: sum=0; for(i=1;i<=n;i++) sum+=i;
                     printf("Sum = %ld\n",sum); break;
            case 2: sum=0; for(i=1;i<=n;i++) sum+=i*i;
                     printf("Sum of squares = %ld\n",sum); break;
            case 3: break;
            default: printf("Invalid choice.\n");
        }
    } while(choice!=3);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang sums.c -o sums
puppie@Anvikas-MacBook-Air Desktop % ./sums
Enter n: 5
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter choice: 1
Sum = 15
```

```
puppie@Anvikas-MacBook-Air Desktop % ./sums
Enter n: 5
Enter choice: 2
Sum of squares = 55
```

Result: The sum program was compiled, executed and tested successfully.

Question 3 - Interest Calculations

Input: principal P, annual rate R (%) and time T (years). **Output:** simple interest, compound amount, compound interest or total payable amount. Annual compounding is used.

Formulae: $SI = PRT/100$; $A = P(1+R/100)^T$; $CI = A - P$; Total payable amount = A.

Algorithm

1. Start. 2. Read P, R and T. 3. Validate inputs. 4. Display menu. 5. Calculate SI when selected. 6. For compound amount, start amount=P and multiply by $(1+R/100)$ T times. 7. $CI = \text{amount} - P$. 8. Total payable=amount. 9. Repeat until Exit. 10. Stop.

Pseudocode

```
READ P,R,T
REPEAT
    DISPLAY menu; READ choice
    SI=P*R*T/100
    amount=P
    FOR i=1 TO T: amount=amount*(1+R/100)
    CI=amount-P
    DISPLAY selected result
UNTIL choice=5
```

Flowchart

START -> Input P,R,T -> Menu -> Calculate selected quantity -> Display -> Menu -> Exit -> STOP.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	10000,10,2; SI	Rs. 2000.00	Rs. 2000.00	Pass
2	same; Amount	Rs. 12100.00	Rs. 12100.00	Pass
3	same; CI	Rs. 2100.00	Rs. 2100.00	Pass
4	same; Total	Rs. 12100.00	Rs. 12100.00	Pass

C Program

```
#include <stdio.h>
int main() {
    double P,R,SI,amount,CI; int T,choice,i;
    printf("Enter principal, rate and time: "); scanf("%lf%lf%d",&P,&R,&T);
    if(P<=0 || R<0 || T<0) { printf("Invalid input.\n"); return 0; }
    do {
        printf("\n1. Simple Interest\n2. Compound Amount\n3. Compound Interest\n4. Total Payable Amount\n5. Exit\nEnter choice: ");
        scanf("%d",&choice);
        SI=P*R*T/100.0; amount=P;
        for(i=1;i<=T;i++) amount*=1.0+R/100.0;
        CI=amount-P;
        switch(choice) {
            case 1: printf("Simple Interest = Rs. %.2f\n",SI); break;
            case 2: printf("Compound Amount = Rs. %.2f\n",amount); break;
            case 3: printf("Compound Interest = Rs. %.2f\n",CI); break;
            case 4: printf("Total Payable Amount = Rs. %.2f\n",amount); break;
            case 5: break;
            default: printf("Invalid choice.\n");
        }
    } while(choice!=5);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang interest.c -o interest
puppie@Anvikas-MacBook-Air Desktop % ./interest
Enter principal, rate and time: 10000 10 2
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter choice: 1
Simple Interest = Rs. 2000.00
```

```
puppie@Anvikas-MacBook-Air Desktop % ./interest
Enter principal, rate and time: 10000 10 2
Enter choice: 2
Compound Amount = Rs. 12100.00
Enter choice: 3
Compound Interest = Rs. 2100.00
Enter choice: 4
Total Payable Amount = Rs. 12100.00
```

Result: The interest program was compiled, executed and tested successfully.